STUDENT COPY — NO SOLVED ANSWERS

Vyom · Is hafte ka print pack

18 Jan – 24 Jan 2027 · Monday–Sunday · NDPS Class 9

Naam: Vyom | Week tasks: 38 | Scheduled tests: 7

Sunday print handover: Sunday, 17 January 2027. Agle Monday se is pack ke daily pages follow karo. Roz tasks tick karo; tests par naam/date aur answers likho. Answer key parent ke paas rahegi.

Din / date	Kaam	Tests / focus
Monday, 18 January 2027	6 tasks	Sab Daily test (10 marks)
Tuesday, 19 January 2027	6 tasks	Sab Daily test (10 marks)
Wednesday, 20 January 2027	6 tasks	Sab Daily test (10 marks)
Thursday, 21 January 2027	6 tasks	Sab Daily test (10 marks)
Friday, 22 January 2027	6 tasks	Sab Daily test (10 marks)
Saturday, 23 January 2027	6 tasks	Sab Daily test (10 marks)
Sunday, 24 January 2027	2 tasks	Maths Full paper (3 hr) (80 marks)

Schedule v4 · 2026-10-12 · Sanskrit coverage. Daily blocks original shifted routine ke mutabik hain; asli weekday date har page par di hai.

School ki subject-wise exam date sheet/paper pattern abhi confirm karna baaki hai. Exam-period sample-section tasks ko official school papers mat samjhein.

Monday, 18 January 2027

Phase 3 · Full revision round

	Samay	Subject	Aaj kya karna hai
☐	6:45–7:10	**Sanskrit**	Revision Sanskrit full-course revision — Path 1 · अहं नमामि · Path 2 · मातृभूमिः · Path 3 · नद्यः: har topic ka book-band oral + written recall, ek mixed exercise, phir source PDFs se check.
☐	2:45–3:30	**Maths**	Revision Pehle homework, phir: Ch 1 · Orienting Yourself: The Use of Coordinates — notes + MCQ + case-based + long Q
☐	3:30–4:00	**Sab**	Test Daily test (10 marks, 30 min): Maths Ch 1 Coordinates · AI ICT Skills · Maths Ch 12 Quadrilaterals
☐	4:00–5:00	**Science**	Revision Full revision: Ch 1 · Exploration: Entering the World of Secondary Science — notes + MCQ + case-based + long Q
☐	5:00–6:00	**SST**	Revision Full revision: Part 1 Ch 1 · Understanding Social Science — notes + MCQ + case-based + long Q
☐	6:00–6:20	**Sab**	Recall Aaj ke 2 chapters ke 5 keywords bolke

Ho gaye: ___ / 6 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Daily Test · 18 Jan

Maths Ch 1 Coordinates · AI ICT Skills · Maths Ch 12 Quadrilaterals

Test ID: D2027-01-18 · Monday, 18 January 2027

Naam: _____ Samay: 30 min Marks: ___ / 10

Slot: 3:30–4:00

Book band. Har answer mein steps/reason likho. Calculator allowed nahi hai.

Daily questions

1. Three vertices of a rectangle are (1, 1), (6, 1) and (6, 4). Find the fourth vertex, the length, breadth and area of the rectangle. **[3]**

Answer space · D2027-01-18 · Q1

2. Differentiate between hardware and software with one example each. **[2]**

Answer space · D2027-01-18 · Q2

3. Write two safe practices while using the internet. **[2]**

Answer space · D2027-01-18 · Q3

4. Prove that in a parallelogram, opposite sides are equal. **[3]**

Answer space · D2027-01-18 · Q4

Teacher/parent: marks answer key ke mutabik; galti C concept / R recall / K careless / T time.

Tuesday, 19 January 2027

Phase 3 · Full revision round

	Samay	Subject	Aaj kya karna hai
☐	6:45–7:10	**Sab**	Recall Formula / keywords notebook — ek subject
☐	2:45–3:30	**Hindi**	Revision Pehle homework, phir: पाठ 1 · दो बैलों की कथा + पाठ 2 · क्या लिखूँ — notes + MCQ + case-based + long Q
☐	3:30–4:00	**Sab**	Test Daily test (10 marks, 30 min): Hindi पाठ 1 · SST Ch 1 · Science Ch 1 Exploration
☐	4:00–5:00	**Maths**	Revision Full revision: Ch 2 · Introduction to Linear Polynomials — notes + MCQ + case-based + long Q
☐	5:00–6:00	**Science**	Revision Full revision: Ch 2 · Cell: The Building Block of Life — notes + MCQ + case-based + long Q
☐	6:00–6:20	**Sab**	Recall Aaj ke 2 chapters ke 5 keywords bolke

Ho gaye: ___ / 6 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Daily Test · 19 Jan

Hindi पाठ 1 · SST Ch 1 · Science Ch 1 Exploration

Test ID: D2027-01-19 · Tuesday, 19 January 2027

Naam: _____ Samay: 30 min Marks: ___ / 10

Slot: 3:30–4:00

Book band. Har answer mein steps/reason likho. Calculator allowed nahi hai.

Daily questions

1. हीरा और मोती का मालिक कौन था? **[1]**

Answer space · D2027-01-19 · Q1

2. कहानी के आधार पर हीरा और मोती की गहरी मित्रता के दो उदाहरण लिखिए। **[2]**

Answer space · D2027-01-19 · Q2

3. 'दो बैलों की कथा' के लेखक कौन हैं? **[1]**

Answer space · D2027-01-19 · Q3

4. Name the four disciplines of Social Science in your Grade 9 textbook. **[1]**

Answer space · D2027-01-19 · Q4

5. Give one reason why studying Social Science helps a citizen. **[1]**

Answer space · D2027-01-19 · Q5

6. Name the four disciplines of Social Science studied in Grade 9. **[1]**

Answer space · D2027-01-19 · Q6

7. Which discipline studies how people make choices about scarce resources? **[1]**

Answer space · D2027-01-19 · Q7

8. Write the steps of the scientific method in order. **[2]**

Answer space · D2027-01-19 · Q8

Teacher/parent: marks answer key ke mutabik; galti C concept / R recall / K careless / T time.

Wednesday, 20 January 2027

Phase 3 · Full revision round

	Samay	Subject	Aaj kya karna hai
☐	6:45–7:10	**Sab**	Recall Formula / keywords notebook — ek subject
☐	2:45–3:30	**SST**	Revision Pehle homework, phir: Part 1 Ch 2 · Shaping of the Earth's Surface — notes + MCQ + case-based + long Q
☐	3:30–4:00	**Sab**	Test Daily test (10 marks, 30 min): SST Ch 2 Earth's surface · Science Ch 2 Cell · Maths Ch 2 Linear Polynomials
☐	4:00–5:00	**English**	Revision Full revision: Unit 1 · How I Taught My Grandmother to Read + Unit 2 · The Pot Maker — notes + MCQ + case-based + long Q
☐	5:00–6:00	**AI**	Revision Full revision: Part A · Communication Skills-I + Part B · AI Reflection, Project Cycle & Ethics — notes + MCQ + case-based + long Q
☐	6:00–6:20	**Sab**	Recall Aaj ke 2 chapters ke 5 keywords bolke

Ho gaye: ___ / 6 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Daily Test · 20 Jan

SST Ch 2 Earth's surface · Science Ch 2 Cell · Maths Ch 2 Linear Polynomials

Test ID: D2027-01-20 · Wednesday, 20 January 2027

Naam: _____ Samay: 30 min Marks: ___ / 10

Slot: 3:30–4:00

Book band. Har answer mein steps/reason likho. Calculator allowed nahi hai.

Daily questions

1. Name the three main layers of the Earth. **[1]**

Answer space · D2027-01-20 · Q1

2. Distinguish between internal (endogenic) and external (exogenic) forces with one example of each. **[2]**

Answer space · D2027-01-20 · Q2

3. Which agent of gradation forms moraines? **[1]**

Answer space · D2027-01-20 · Q3

4. Which organelle is the site of cellular respiration? **[1]**

Answer space · D2027-01-20 · Q4

5. Name two structures present in a plant cell but absent in an animal cell. Write the function of any one. **[2]**

Answer space · D2027-01-20 · Q5

6. Write the slope and y-intercept of the line $y = 5x - 2$. **[1]**

Answer space · D2027-01-20 · Q6

7. A linear polynomial p(x) has slope –3 and y-intercept 4. Write p(x) and find p(2). **[2]**

Answer space · D2027-01-20 · Q7

Teacher/parent: marks answer key ke mutabik; galti C concept / R recall / K careless / T time.

Thursday, 21 January 2027

Phase 3 · Full revision round

	Samay	Subject	Aaj kya karna hai
☐	6:45–7:10	**Sab**	Recall Formula / keywords notebook — ek subject
☐	2:45–3:30	**Maths**	Revision Pehle homework, phir: Ch 3 · The World of Numbers — notes + MCQ + case-based + long Q
☐	3:30–4:00	**Sab**	Test Daily test (10 marks, 30 min): Maths Ch 3 World of Numbers · AI Communication Skills · English Unit 1
☐	4:00–5:00	**Science**	Revision Full revision: Ch 3 · Tissues in Action — notes + MCQ + case-based + long Q
☐	5:00–6:00	**SST**	Revision Full revision: Part 1 Ch 3 · Atmosphere and Climate — notes + MCQ + case-based + long Q
☐	6:00–6:20	**Sab**	Recall Aaj ke 2 chapters ke 5 keywords bolke

Ho gaye: ___ / 6 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Daily Test · 21 Jan

Maths Ch 3 World of Numbers · AI Communication Skills · English Unit 1

Test ID: D2027-01-21 · Thursday, 21 January 2027

Naam: _____ Samay: 30 min Marks: ___ / 10

Slot: 3:30–4:00

Book band. Har answer mein steps/reason likho. Calculator allowed nahi hai.

Daily questions

1. Simplify $(\sqrt{5} + \sqrt{2})^2$ and $(\sqrt{5} + \sqrt{2})(\sqrt{5} - \sqrt{2})$. Is each result rational or irrational? **[3]**

Answer space · D2027-01-21 · Q1

2. Write the stages of the communication cycle in the correct order. **[1]**

Answer space · D2027-01-21 · Q2

3. Give one example of non-verbal communication. **[1]**

Answer space · D2027-01-21 · Q3

4. Write any two of the 7 Cs of communication with their meaning. **[2]**

Answer space · D2027-01-21 · Q4

5. Which story did the grandmother want to read herself in “How I Taught My Grandmother to Read”? **[1]**

Answer space · D2027-01-21 · Q5

6. Why was the grandmother in tears when the narrator came back from the wedding? **[2]**

Answer space · D2027-01-21 · Q6

Teacher/parent: marks answer key ke mutabik; galti C concept / R recall / K careless / T time.

Friday, 22 January 2027

Phase 3 · Full revision round

	Samay	Subject	Aaj kya karna hai
☐	6:45–7:10	**Sab**	Recall Formula / keywords notebook — ek subject
☐	2:45–3:30	**Hindi**	Revision Pehle homework, phir: पाठ 3 · संवादहीन + पाठ 4 · ऐसी भी बातें होती हैं — notes + MCQ + case-based + long Q
☐	3:30–4:00	**Sab**	Test Daily test (10 marks, 30 min): Hindi पाठ 3 संवादहीन · SST Ch 3 Atmosphere layers · Science Ch 3 Plant tissues
☐	4:00–5:00	**Maths**	Revision Full revision: Ch 4 · Exploring Algebraic Identities — notes + MCQ + case-based + long Q
☐	5:00–6:00	**Science**	Revision Full revision: Ch 4 · Describing Motion Around Us — notes + MCQ + case-based + long Q
☐	6:00–6:20	**Sab**	Recall Aaj ke 2 chapters ke 5 keywords bolke

Ho gaye: ___ / 6 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Daily Test · 22 Jan

Hindi पाठ 3 संवादहीन · SST Ch 3 Atmosphere layers · Science Ch 3 Plant tissues

Test ID: D2027-01-22 · Friday, 22 January 2027

Naam: _____ Samay: 30 min Marks: ___ / 10

Slot: 3:30–4:00

Book band. Har answer mein steps/reason likho. Calculator allowed nahi hai.

Daily questions

1. 'संवादहीन' कहानी के लेखक कौन हैं? **[1]**

Answer space · D2027-01-22 · Q1

2. कहानी के दो मुख्य पात्र कौन हैं? **[1]**

Answer space · D2027-01-22 · Q2

3. ताई के लिए मिट्ठू केवल एक तोता क्यों नहीं था? **[2]**

Answer space · D2027-01-22 · Q3

4. About what percentage of the atmosphere is oxygen? **[1]**

Answer space · D2027-01-22 · Q4

5. Name the layer in which almost all weather phenomena occur. Write two features of this layer. **[2]**

Answer space · D2027-01-22 · Q5

6. Which plant tissue carries water from the roots to the leaves? **[1]**

Answer space · D2027-01-22 · Q6

7. Write two features of meristematic cells. Name any two types of meristem by location. **[2]**

Answer space · D2027-01-22 · Q7

Teacher/parent: marks answer key ke mutabik; galti C concept / R recall / K careless / T time.

Saturday, 23 January 2027

Phase 3 · Full revision round

	Samay	Subject	Aaj kya karna hai
☐	6:45–7:10	**Sab**	Recall Formula / keywords notebook — ek subject
☐	2:45–3:30	**SST**	Revision Pehle homework, phir: Part 1 Ch 4 · Early Humans and Beginning of Civilisation — notes + MCQ + case-based + long Q
☐	3:30–4:00	**Sab**	Test Daily test (10 marks, 30 min): SST Ch 4 Early humans · Science Ch 4 Motion · Maths Ch 4 Identities
☐	4:00–5:00	**English**	Revision Full revision: Unit 3 · Winds of Change + Unit 4 · Vitamin-M — notes + MCQ + case-based + long Q
☐	5:00–6:00	**AI**	Revision Full revision: Part A · Self-Management Skills-I + Part A · ICT Skills-I — notes + MCQ + case-based + long Q
☐	6:00–6:20	**Sab**	Recall Aaj ke 2 chapters ke 5 keywords bolke

Ho gaye: ___ / 6 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Daily Test · 23 Jan

SST Ch 4 Early humans · Science Ch 4 Motion · Maths Ch 4 Identities

Test ID: D2027-01-23 · Saturday, 23 January 2027

Naam: _____ Samay: 30 min Marks: ___ / 10

Slot: 3:30–4:00

Book band. Har answer mein steps/reason likho. Calculator allowed nahi hai.

Daily questions

1. In which state are the Bhimbetka rock shelters located? **[1]**

Answer space · D2027-01-23 · Q1

2. Write two important changes in human life during the Neolithic period. **[2]**

Answer space · D2027-01-23 · Q2

3. A train moving at 25 m s^{-1} applies brakes and stops in 10 s with uniform retardation. Find the retardation and the distance travelled before stopping. **[3]**

Answer space · D2027-01-23 · Q3

4. Expand $(a + 2b - c)^2$. **[2]**

Answer space · D2027-01-23 · Q4

5. If $a + b = 7$ and $ab = 10$, find $a^2 + b^2$. **[2]**

Answer space · D2027-01-23 · Q5

Teacher/parent: marks answer key ke mutabik; galti C concept / R recall / K careless / T time.

Sunday, 24 January 2027

Phase 3 · Full revision round

	Samay	Subject	Aaj kya karna hai
☐	9:00–12:00	**Maths**	Test Maths full paper — school pattern, 3 hr timed
☐	Shaam (60 min)	**Maths**	Correction Checking + har galti ka reason; Mistakes notebook

Ho gaye: ___ / 2 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Maths · Full Paper 1 (Annual pattern) · 24 Jan

Class 9 · Ganita Manjari Part I + II — poora syllabus (Ch 1–14)

Test ID: T2027-01-24 · Sunday, 24 January 2027

Naam: _____ Samay: 180 min Marks: ___ / 80

Slot: 9:00–12:00

Paper mein 38 prashn aur 5 section hain. Saare prashn compulsory hain; Section D mein 2 prashnon mein internal choice hai.
Section A: 1 mark · B: 2 marks · C: 3 marks · D: 5 marks · E: case-based 4 marks.
Jahan zaroori ho π = 22/7 lo. Har step likho.
3 ghante ka timer lagao; pehle 15 minute sirf paper padho.

Section A · MCQ + Assertion-Reason · 1 mark each

1. The point (0, –7) lies: **[1]**

(a) on the x-axis (b) on the y-axis
(c) in quadrant III (d) in quadrant IV

Answer space · T2027-01-24 · Q1

2. The x-coordinate (abscissa) of the point (–4, 9) is: **[1]**

(a) 9 (b) –4
(c) 4 (d) –9

Answer space · T2027-01-24 · Q2

3. Which of the following is a linear polynomial? **[1]**

(a) $x^2 + 1$ (b) 3 – 2x
(c) x^3 (d) 1/x

Answer space · T2027-01-24 · Q3

4. For p(x) = 3x – 6, the value of x for which p(x) = 0 is: **[1]**

(a) –2 (b) 3
(c) 2 (d) 6

Answer space · T2027-01-24 · Q4

5. 0.333... written as p/q is: **[1]**

(a) 3/10 (b) 1/3

(c) 33/100 (d) 3/100

Answer space · T2027-01-24 · Q5

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6. $\sqrt{12} \times \sqrt{3} =$ **[1]**

(a) 6 (b) $\sqrt{15}$

(c) 36 (d) $3\sqrt{2}$

Answer space · T2027-01-24 · Q6

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7. $(x + 2)(x + 3) =$ **[1]**

(a) $x^2 + 6$ (b) $x^2 + 5x + 6$

(c) $x^2 + 6x + 5$ (d) $x^2 + 5x + 5$

Answer space · T2027-01-24 · Q7

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8. Using an identity, $105 \times 95 =$ **[1]**

(a) 9975 (b) 10025

(c) 9925 (d) 9995

Answer space · T2027-01-24 · Q8

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9. An arc subtends 100° at the centre of a circle. The angle it subtends at a point on the remaining part of the circle is: **[1]**

(a) 100°

(b) 200°

(c) 50°

(d) 80°

Answer space · T2027-01-24 · Q9

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10. The circumference of a circle of diameter 14 cm is: **[1]**

(a) 22 cm (b) 44 cm

(c) 88 cm (d) 154 cm

Answer space · T2027-01-24 · Q10

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11. Two coins are tossed. The probability of getting no head is: **[1]**

(a) 1/2 (b) 1/4

(c) 3/4 (d) 0

Answer space · T2027-01-24 · Q11

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12. The n-th term of the GP 3, 9, 27, ... is: **[1]**

(a) 3n (b) 3^n

(c) n^3 (d) 3^{n-1}

Answer space · T2027-01-24 · Q12

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13. 1 + 2 + 3 + ... + 20 = **[1]**

(a) 200 (b) 210

(c) 220 (d) 190

Answer space · T2027-01-24 · Q13

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14. A counterexample to "All prime numbers are odd" is: **[1]**

(a) 3 (b) 9

(c) 2 (d) 1

Answer space · T2027-01-24 · Q14

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15. The weighted mean of 10, 20, 30 with weights 1, 2, 3 is about: **[1]**

(a) 20 (b) 23.33

(c) 25 (d) 30

Answer space · T2027-01-24 · Q15

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16. Using Euclid's algorithm, gcd(48, 18) = **[1]**

(a) 3 (b) 6

(c) 9 (d) 18

Answer space · T2027-01-24 · Q16

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17. The diagonals of a rhombus: **[1]**

(a) are always equal (b) bisect each other at right angles

(c) do not bisect each other (d) are parallel

Answer space · T2027-01-24 · Q17

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18. Which line is parallel to $y = 3x + 1$? **[1]**

(a) $y = -3x + 1$ (b) $y = x + 3$

(c) $y = 3x - 5$ (d) $y = (1/3)x$

Answer space · T2027-01-24 · Q18

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19. **A:** Every rational number is an integer. **[1]**
R: Every integer n can be written as n/1.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Answer space · T2027-01-24 · Q19

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20. **A:** The pair $2x + 3y = 5$ and $4x + 6y = 7$ has no solution. **[1]**
R: Two lines are parallel when $a_1/a_2 = b_1/b_2 \neq c_1/c_2$.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Answer space · T2027-01-24 · Q20

Section B · Very Short Answer · 2 marks each

21. Find k if $x = 2, y = -1$ is a solution of $3x + ky = 8$. **[2]**

Answer space · T2027-01-24 · Q21

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22. Find the curved surface area of a cone of radius 7 cm and slant height 10 cm. **[2]**

Answer space · T2027-01-24 · Q22

23. Find the 12th term of the AP –5, –1, 3, 7, ... **[2]**

Answer space · T2027-01-24 · Q23

24. A die is thrown 60 times and a six comes up 12 times. Find the experimental probability of a six and compare it with the theoretical probability. **[2]**

Answer space · T2027-01-24 · Q24

25. Simplify $(x^2 – 4)/(x^2 + x – 6)$. **[2]**

Answer space · T2027-01-24 · Q25

Section C · Short Answer · 3 marks each

26. Express 0.2353535... (35 repeating) in the form p/q. **[3]**

Answer space · T2027-01-24 · Q26

27. In quadrilateral ABCD, AB = 3 cm, BC = 4 cm, CD = 4 cm, DA = 5 cm and diagonal AC = 5 cm. Find its area. **[3]**

Answer space · T2027-01-24 · Q27

28. A chord of length 24 cm is at a distance of 5 cm from the centre of a circle. Find the radius. Also find the length of a chord at a distance of 12 cm from the centre. **[3]**

Answer space · T2027-01-24 · Q28

29. A bag has 2 white and 3 black balls. A ball is drawn, put back, and a second ball is drawn. Find P(both black) and P(balls of different colours). **[3]**

Answer space · T2027-01-24 · Q29

30. In parallelogram ABCD, the bisectors of $\angle A$ and $\angle B$ meet at P. Prove that $\angle APB = 90^\circ$. **[3]**

Answer space · T2027-01-24 · Q30

31. A road roller has diameter 84 cm and length 120 cm. It takes 500 complete revolutions to level a playground. Find the area of the playground in m^2. **[3]**

Answer space · T2027-01-24 · Q31

Section D · Long Answer · 5 marks each

32. Draw the graphs of $2x + y = 8$ and $x - y = 1$. (i) Find their point of intersection. (ii) Find the vertices and area of the triangle formed by these lines and the x-axis. **[5]**

ATHVA / OR

If 2 is added to both the numerator and denominator of a fraction, it becomes 9/11. If 3 is added to both, it becomes 5/6. Find the fraction.

Answer space · T2027-01-24 · Q32

33. (i) If $x - y = 3$ and $xy = 10$, find $x^3 - y^3$. (ii) Factorise $27a^3 + 64b^3$. (iii) Evaluate 103^3 using an identity. **[5]**

Answer space · T2027-01-24 · Q33

34. Prove that the angle subtended by an arc at the centre of a circle is double the angle subtended by it at any point on the remaining part of the circle. **[5]**

ATHVA / OR

Prove that equal chords of a circle subtend equal angles at the centre.

Answer space · T2027-01-24 · Q34

35. A toy is a hemisphere of radius 3.5 cm with a cone of the same radius on top. The total height of the toy is 15.5 cm. Find (i) the slant height of the cone (ii) the total surface area of the toy (iii) its volume. **[5]**

Answer space · T2027-01-24 · Q35

Section E · Case-Based · 4 marks each

36. **[4]**

> A school auditorium has 20 seats in the first row, 22 in the second, 24 in the third, and so on.

(i) How many seats are there in the 10th row? [1]
(ii) Which row has 60 seats? [1]
(iii) By pairing the first and the last of the first 10 rows (row 1 + row 10, row 2 + row 9, ...), find the total number of seats in the first 10 rows. [2]

Answer space · T2027-01-24 · Q36

37. **[4]**

> Two branches of a stationery shop sold pens in January and February: Branch A sold 120 and 80; Branch B sold 60 and 140.

(i) What were the total pen sales in January? [1]
(ii) In a 100% stacked bar chart for January, what percentage of the bar would Branch A take? [1]
(iii) Branch A sold its 200 pens at ₹10 each and Branch B its 200 pens at ₹12 each. Find the average selling price of all 400 pens. If Branch B had sold 300 pens at ₹12, what would the average be? [2]

Answer space · T2027-01-24 · Q37

38. **[4]**

> Riya finds all divisors of 36 with this algorithm: for each j from 1 to 36, if j divides 36, add j to the list.

(i) Write the list she gets. [1]
(ii) How many numbers does she check? [1]
(iii) Her brother checks only j = 1 to 6 and, whenever j divides 36, writes both j and 36/j. How many checks does he make, and why does his method still find every divisor? [2]

Answer space · T2027-01-24 · Q38

Teacher/parent: marks answer key ke mutabik; galti C concept / R recall / K careless / T time.

Extra answer sheet · 18 Jan – 24 Jan

Naam: _____ Date: _____ Subject: _____

Test ID / question number: _____

Diagram / graph aur Sunday review

Test ID / question number: _____



Is hafte ki 3 galtiyan aur sudhaar:

Week tasks: ___ / 38

Parent sign: _____

Agla hafta: sabse zaroori weak topic / pending kaam: